Overview

In this rebuttal supplement, we provide comprehensive experimental evidence addressing concerns on **scalability, numerical precision, and convergence stability** of Phase Gradient Flow (PGF). Specifically, we include: 1) **Constant memory scaling** $O(1)$ analysis showing up to **17× memory reduction** ($L = 4096$, 8 layers); 2) **Component-wise gradient exactness** with errors below 10^{-5}; 3) **End-to-end convergence** on WikiText-2 matching Autograd; and 4) **Extreme sequence length** (1M+) training with stable **1.1 GB** memory footprint.

1. Efficiency & Scalability: Constant Memory $O(1)$

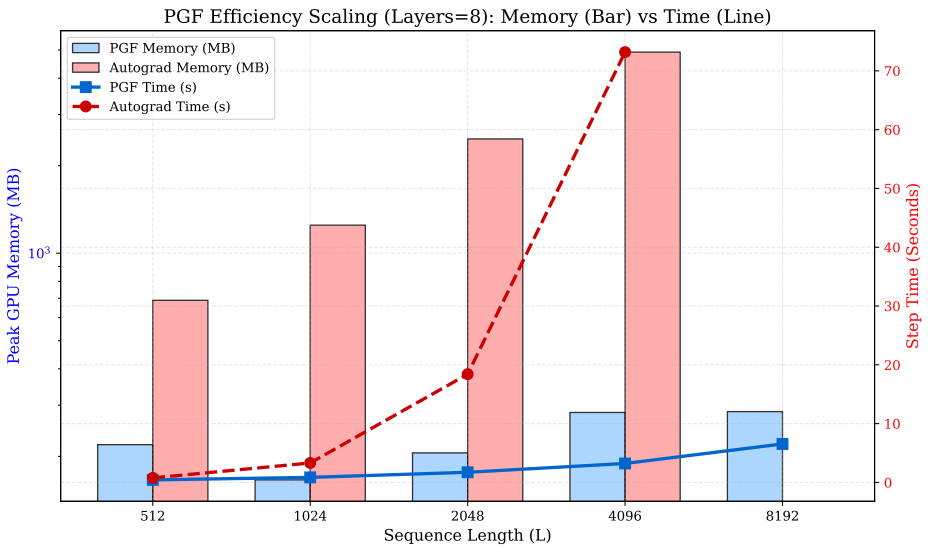
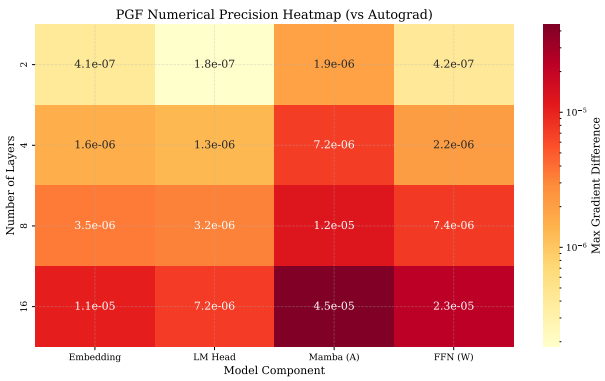


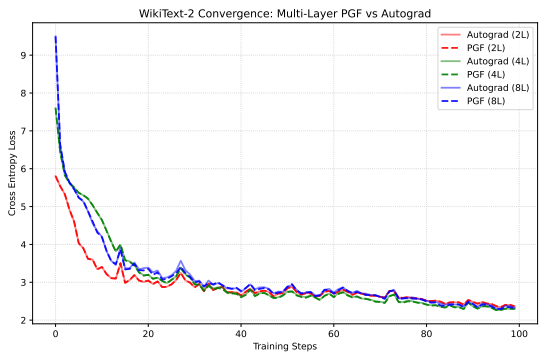
Figure 1: **Memory and Time Scaling (8 Layers).** PGF maintains a near-constant memory footprint (∼**285 MB**) across sequence lengths up to 8192, while Autograd memory grows linearly and triggers **OOM** at $L = 8192$. At $L = 4096$, PGF reduces peak memory from **4.9 GB** to **283 MB**.

2. Numerical Exactness & Convergence

We demonstrate that PGF is mathematically equivalent to standard backpropagation, with negligible numerical differences.



(a) **Gradient Heatmap.** Max L_∞ error per component. Numerical errors are consistently below 10^{-5} across all model components (Embedding, LM Head, Mamba, FFN), ensuring long-term training stability.



(b) **WikiText-2 Convergence.** PGF and Autograd loss curves are indistinguishable, with both reaching a final loss of approximately **2.3** within 100 steps, confirming optimization integrity.

Figure 2: Numerical and Functional Consistency Analysis.

3. Extreme Length Capability (L=1M+)

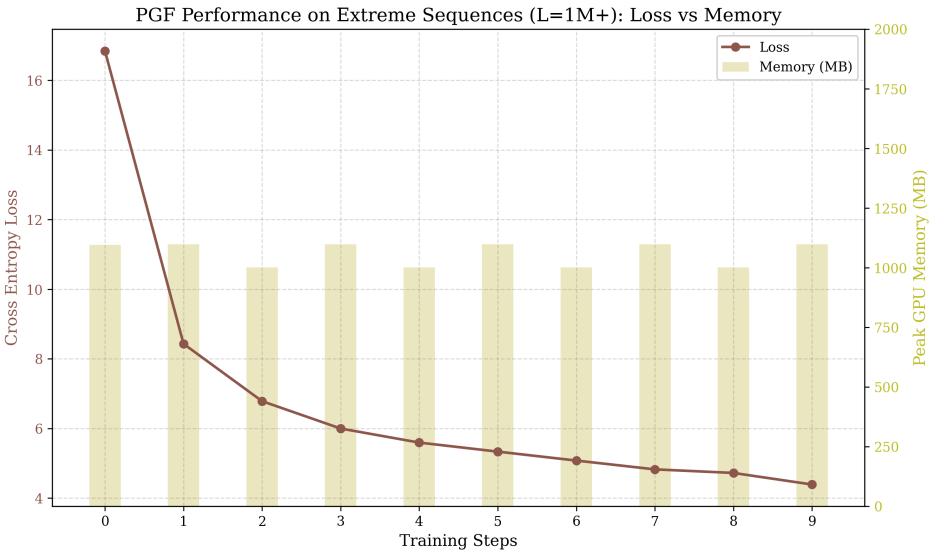


Figure 3: **Extreme Sequence Length Analysis.** PGF successfully trains on ultra-long sequences (1M+ tokens) with a stable memory consumption of **1.1 GB**. The loss rapidly decreases from **16.8** to **4.3**, demonstrating successful optimization in regimes entirely inaccessible to standard Autograd-based Mamba implementations.